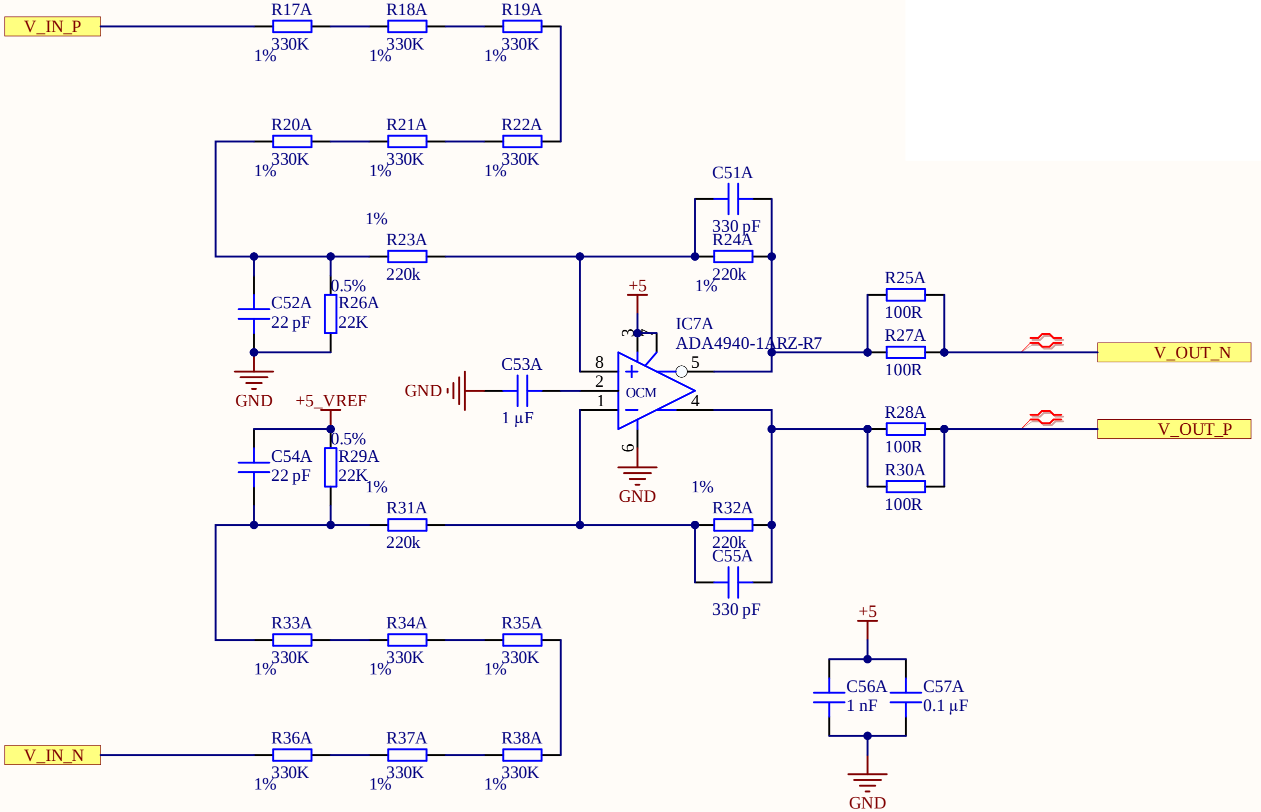
Vin: 0V - 900V
Vout: -4.5V - 4.5V
-3dB @2.2kHz
G = 0,01



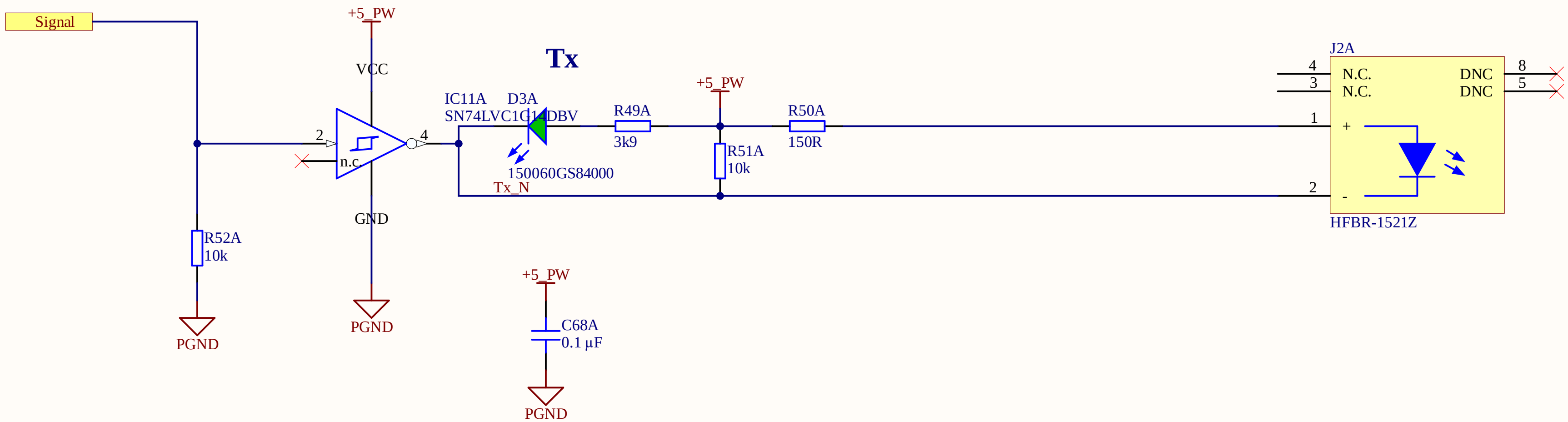
Title Voltage_Measurement.SchDoc		UltraZohm
Revision: Rev01	Design Engineer: Veit Starost	www.ultrazohm.com
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB		Date: 08.03.2023
		Sheet .1 of 25

Poor Mans Table of Truth for Tx

Transmitter LED	Node Tx_N	LED D3	Signal
OFF	HIGH	OFF	LOW
ON	LOW	ON	HIGH

Poor Mans Table of Truth for Rx

Receiver LED	Node RxTx	LED D2
OFF	HIGH	OFF
ON	LOW	ON



Title Optical_TX.SchDoc		UltraZohm
Revision: Rev01	Design Engineer: Veit Starost	www.ultrazohm.com
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB		Date: 08.03.2023 Sheet .1 of 25

Poor Mans Table of Truth for Tx

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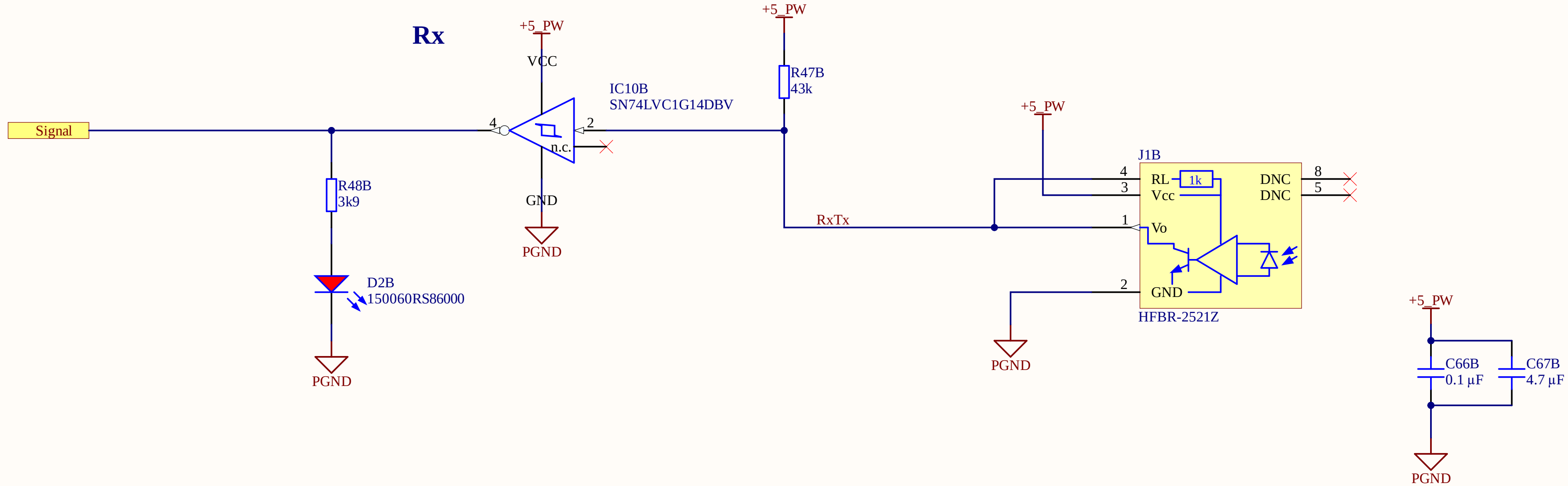
OFF	HIGH	OFF	LOW
ON	LOW	ON	HIGH

Poor Mans Table of Truth for Rx

Receiver LED	Node RxTx	LED D2
OFF	HIGH	OFF
ON	LOW	ON



Title Optical_RX.SchDoc	*UltraZohm*
Revision: Rev01 · Design Engineer: Veit Starost	www.ultrazohm.com
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB	Date: 08.03.2023 Sheet .1 of 25

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Poor Mans Table of Truth for Rx

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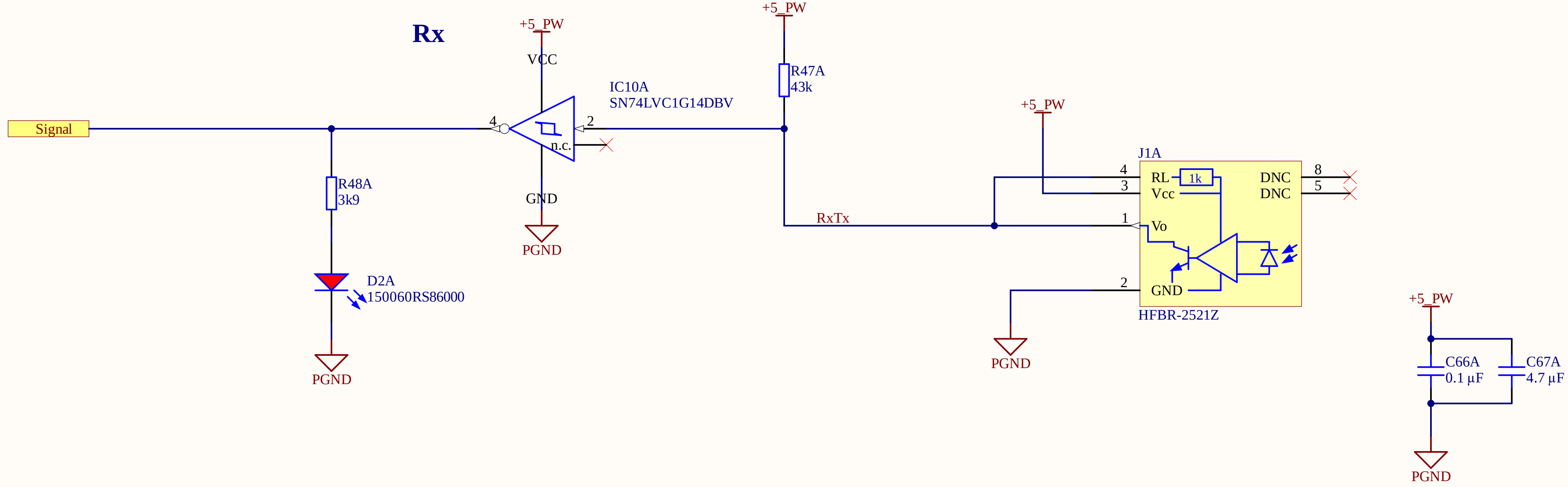
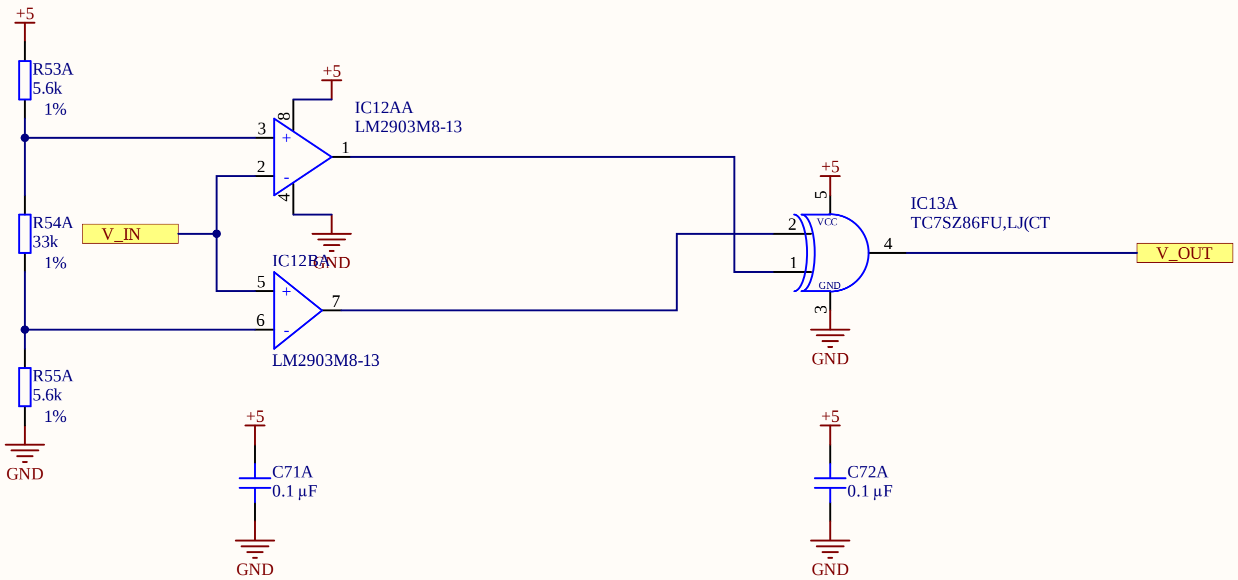
OFF	HIGH	OFF
ON	LOW	ON



Title Optical_RX.SchDoc	UltraZohm
Revision: Rev01 — Design Engineer: Veit Starost	www.ultrazohm.com
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB	Date: 08.03.2023 Sheet .1 of 25



Window comparator circuit
Threshold calculation acc to screenshot
XOR to select areas where VIN is out of the window

VH 4.35V 45A
VL 0.65V -45A

2 Theory of Operation

Figure 2 depicts a more detailed schematic for this reference design.

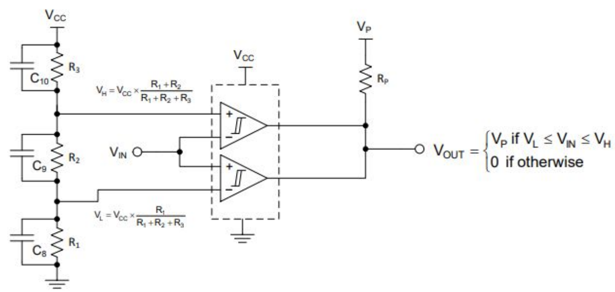


Figure 2: Detailed Window Comparator Schematic

A reference voltage, V_{CC}, is divided down by resistors R_1-R_3. The two node voltages, V_H and V_L, define the upper window voltage and lower window voltage, respectively. When the input voltage is between V_H and V_L, the output is 'high', or V_P. When outside the window voltage, the output is pulled down to 0 V.

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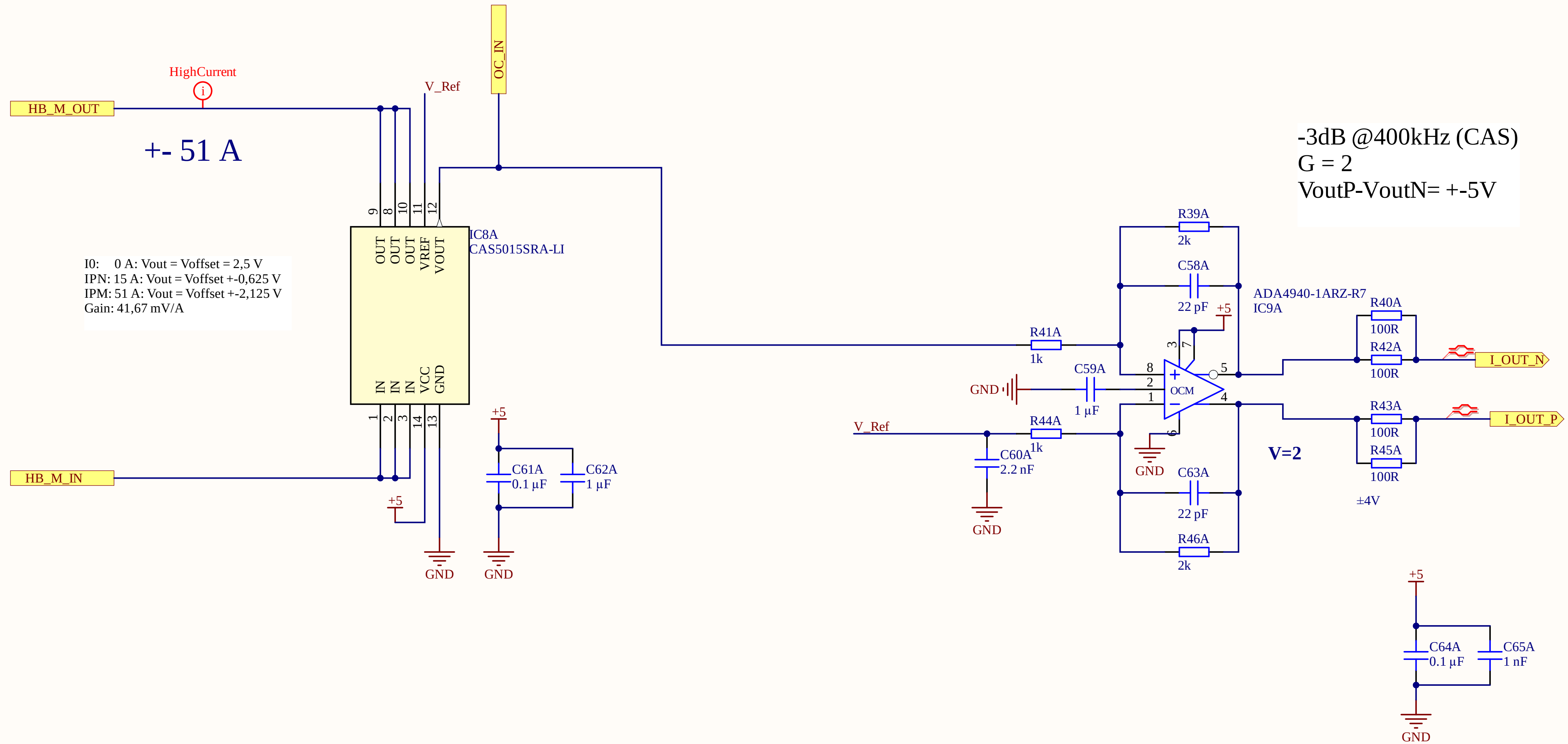
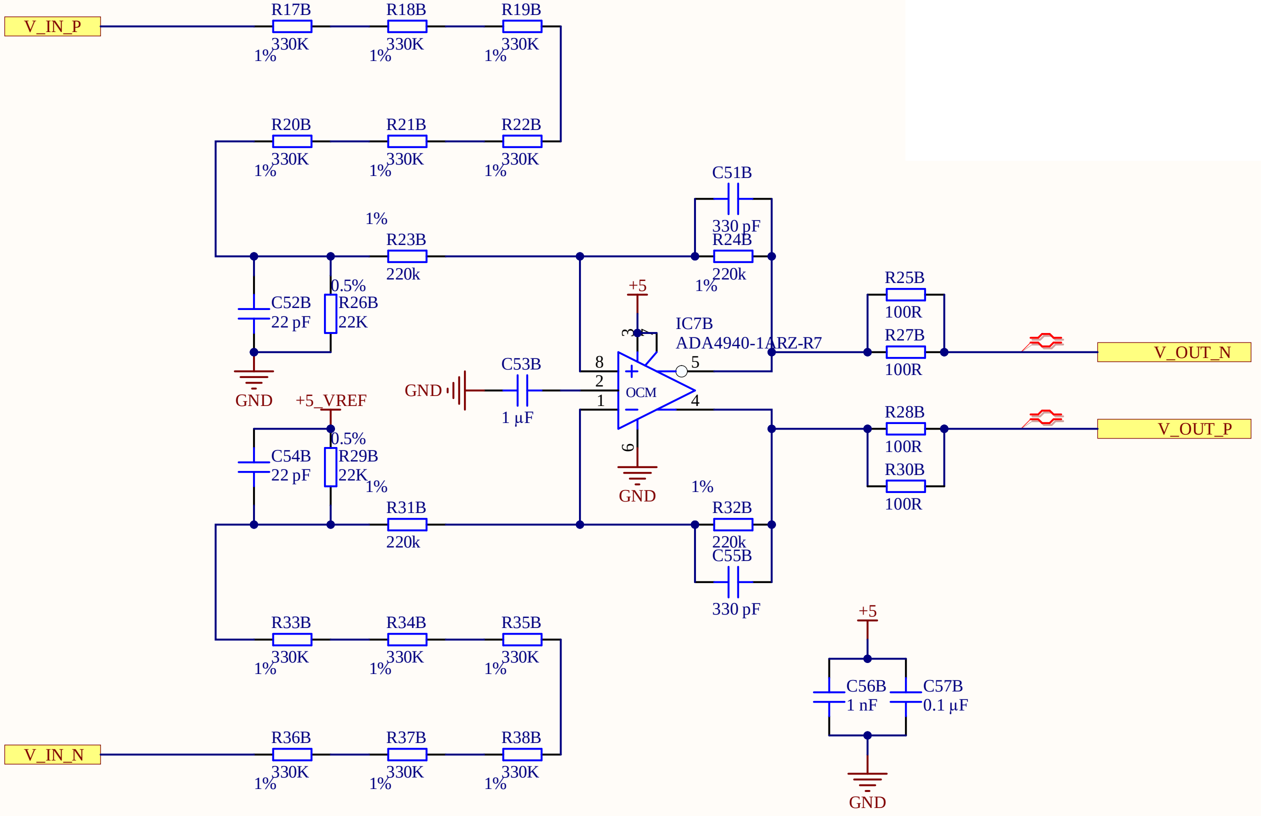
$$V_L = V_{CC} \times \frac{R_1}{R_1 + R_2 + R_3} \quad (2)$$

Title OC_Trigger.SchDoc		UltraZohm www.ultrazohm.com
Revision: Rev01	Design Engineer: Veit Starost	
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB		Date: 08.03.2023 Sheet .1 of 25



Title Current_Measurement_CAS5000.SchDoc		*UltraZohm* www.ultrazohm.com
Revision: Rev01	Design Engineer: Veit Starost	
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB		Date: 08.03.2023 Sheet .1 of 25

Vin: 0V - 900V
Vout: -4.5V - 4.5V
-3dB @2.2kHz
G = 0,01



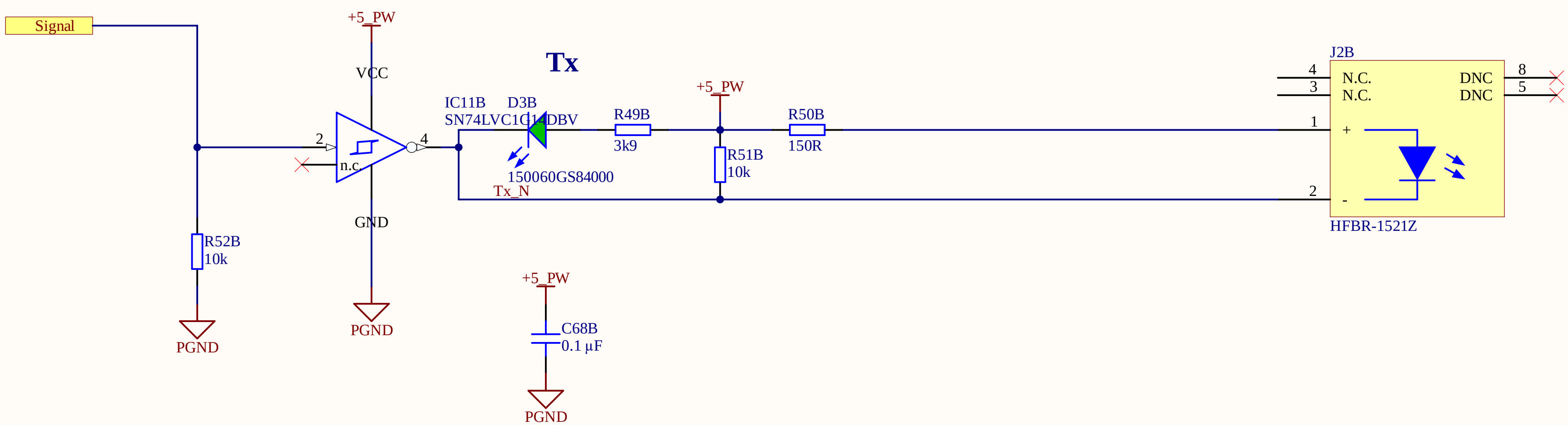
Title Voltage_Measurement.SchDoc	UltraZohm www.ultrazohm.com
Revision: Rev01 Design Engineer: Veit Starost	
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB	Date: 08.03.2023 Sheet .2 of 25

Poor Mans Table of Truth for Tx

Transmitter LED	Node Tx_N	LED D3	Signal
OFF	HIGH	OFF	LOW
ON	LOW	ON	HIGH

Poor Mans Table of Truth for Rx

Receiver LED	Node RxTx	LED D2
OFF	HIGH	OFF
ON	LOW	ON



Title Optical_TX.SchDoc	**UltraZohm**
Revision: Rev01 · Design Engineer: Veit Starost	www.ultrazohm.com
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB	Date: 08.03.2023 · Sheet .2 of 25

Poor Mans Table of Truth for Tx

Transmitter LED	Node Tx_N	LED D3	Signal
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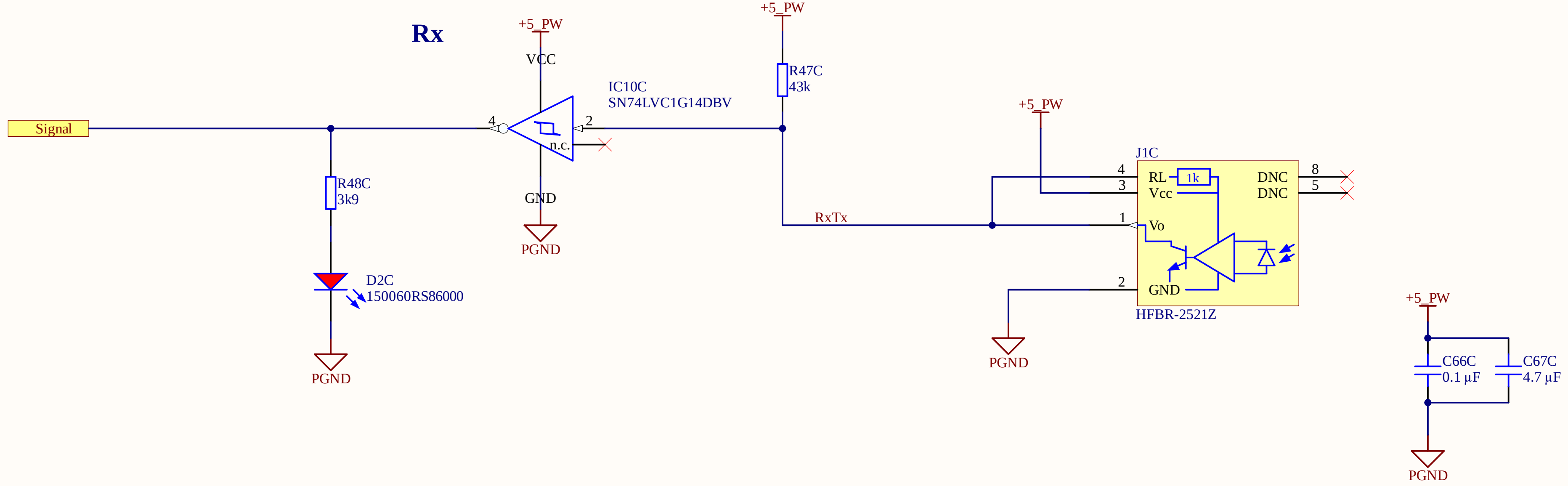
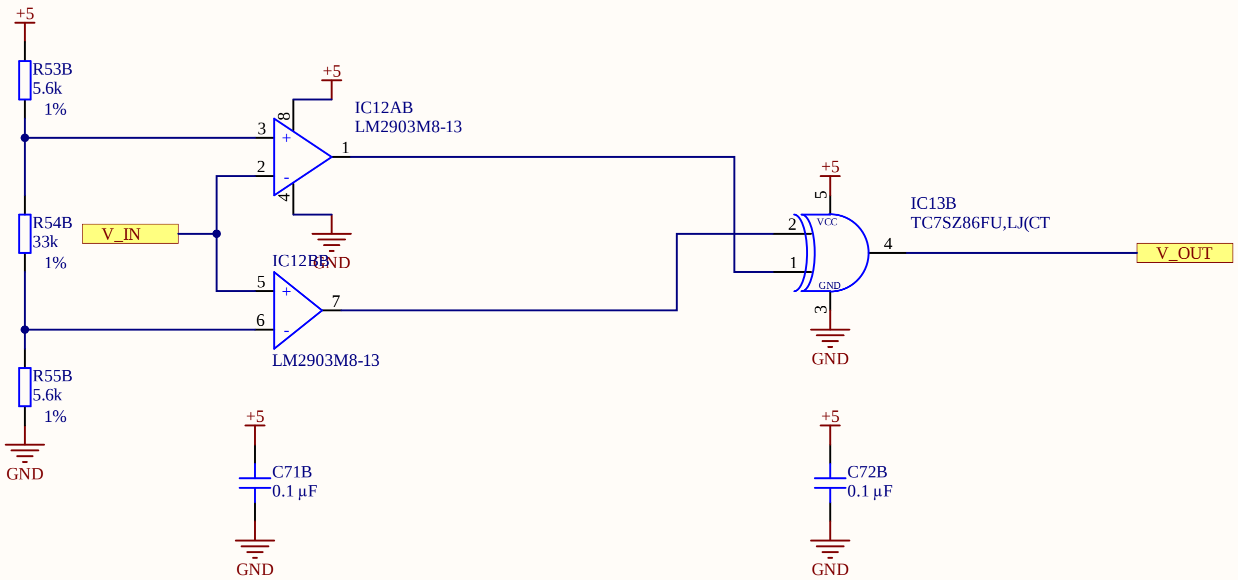
Poor Mans Table of Truth for Rx

Receiver LED	Node RxTx	LED D2
OFF	HIGH	OFF
ON	LOW	ON



Title Optical_RX.SchDoc	UltraZohm
Revision: Rev01	Design Engineer: Veit Starost
www.ultrazohm.com	
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB	Date: 08.03.2023
	Sheet .2 of 25



Window comparator circuit
Threshold calculation acc to screenshot
XOR to select areas where VIN is out of the window

VH 4.35V 45A
VL 0.65V -45A

2 Theory of Operation

Figure 2 depicts a more detailed schematic for this reference design.

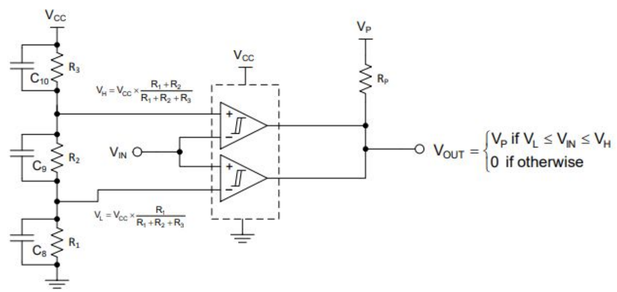


Figure 2: Detailed Window Comparator Schematic

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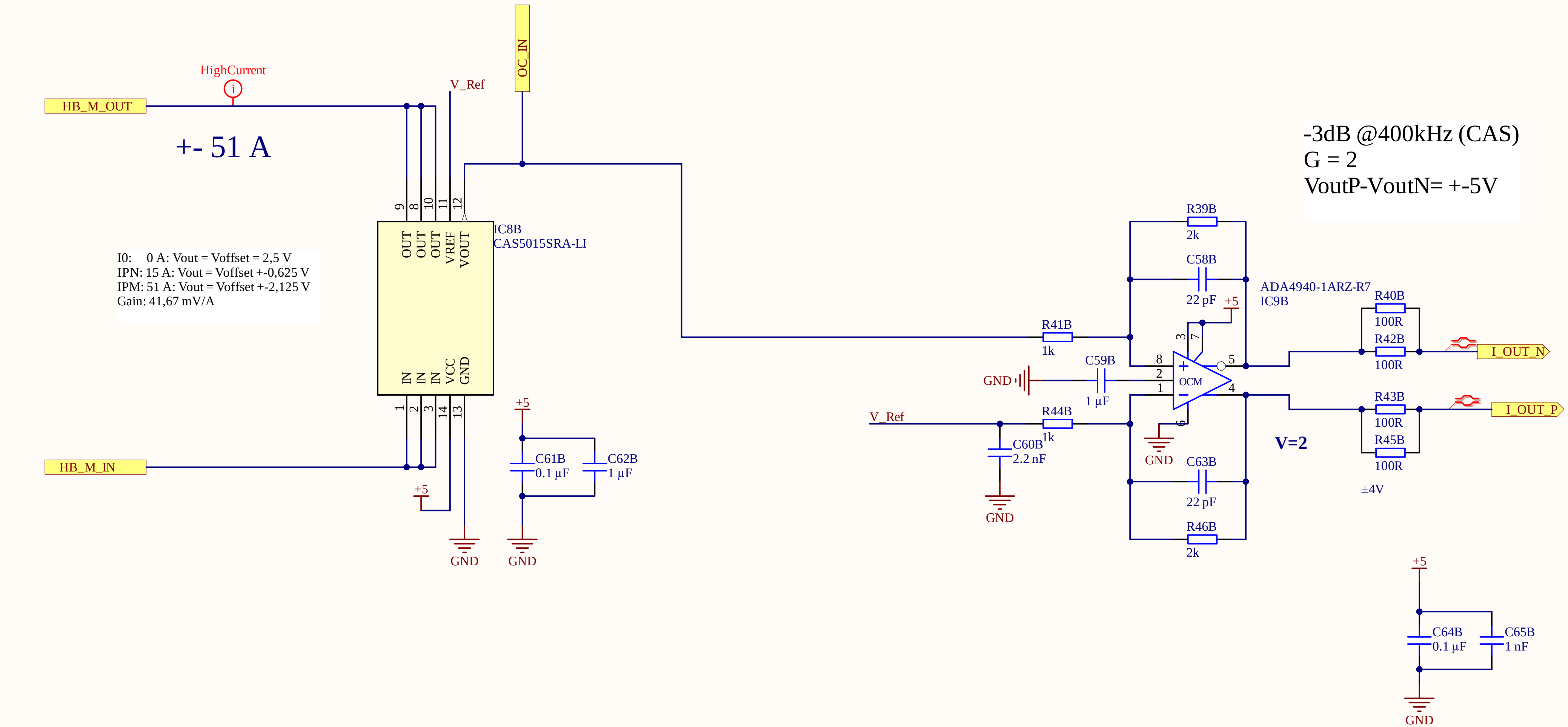
Equations (1) and (2) define V_H and V_L, respectively.

$$V_H = V_{CC} \times \frac{R_1+R_2}{R_1+R_2+R_3} \tag{1}$$

$$V_L = V_{CC} \times \frac{R_1}{R_1+R_2+R_3} \tag{2}$$

Title OC_Trigger.SchDoc	UltraZohm
Revision: Rev01 / Design Engineer: Veit Starost	www.ultrazohm.com
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB	Date: 08.03.2023 / Sheet .2 of 25



Title	Current_Measurement_CAS5000.SchDoc
Revision: Rev01	Design Engineer: Veit Starost
Project:	UZ_PER_SiC_inverter_828xcc.PrjPCB

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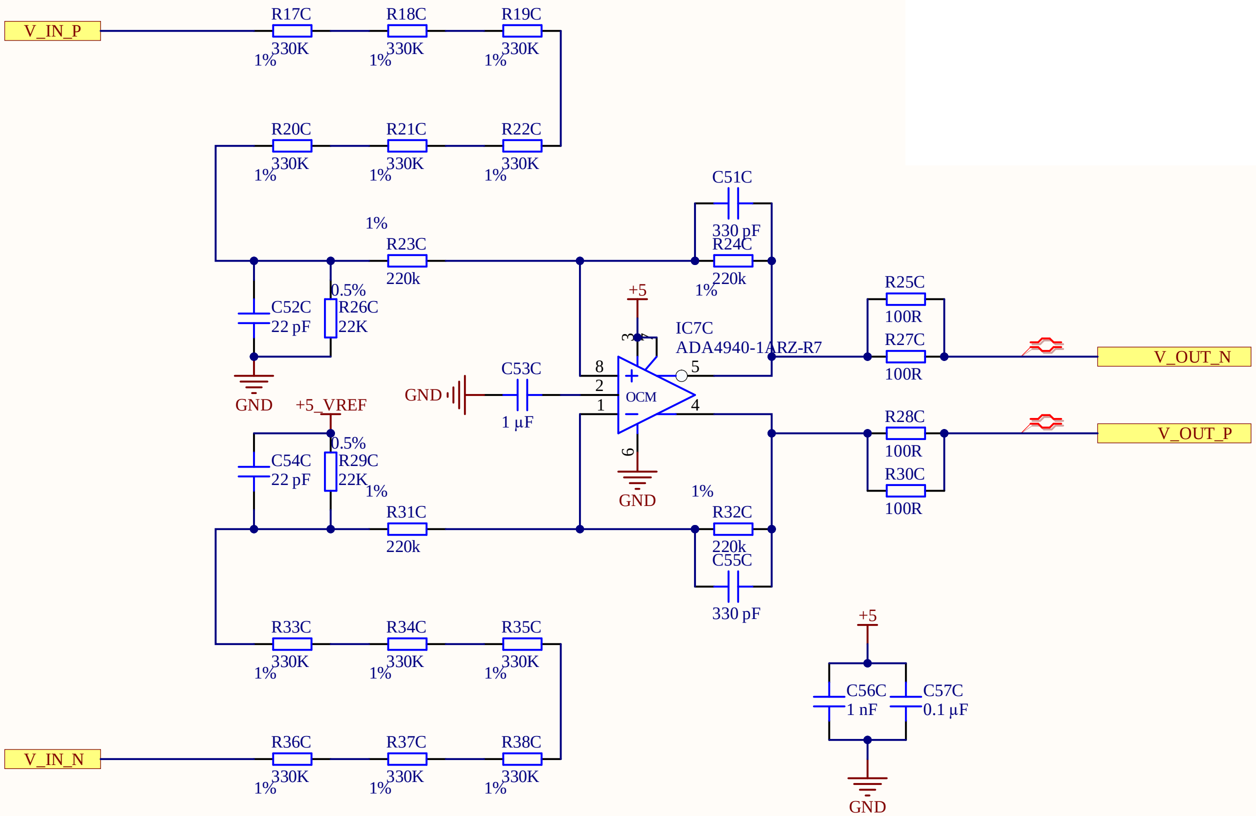
www.ultrazohm.com

Date: 08.03.2023
Sheet .2 of 25

Vin: 0V - 900V
Vout: -4.5V - 4.5V
-3dB @2.2kHz
G = 0,01



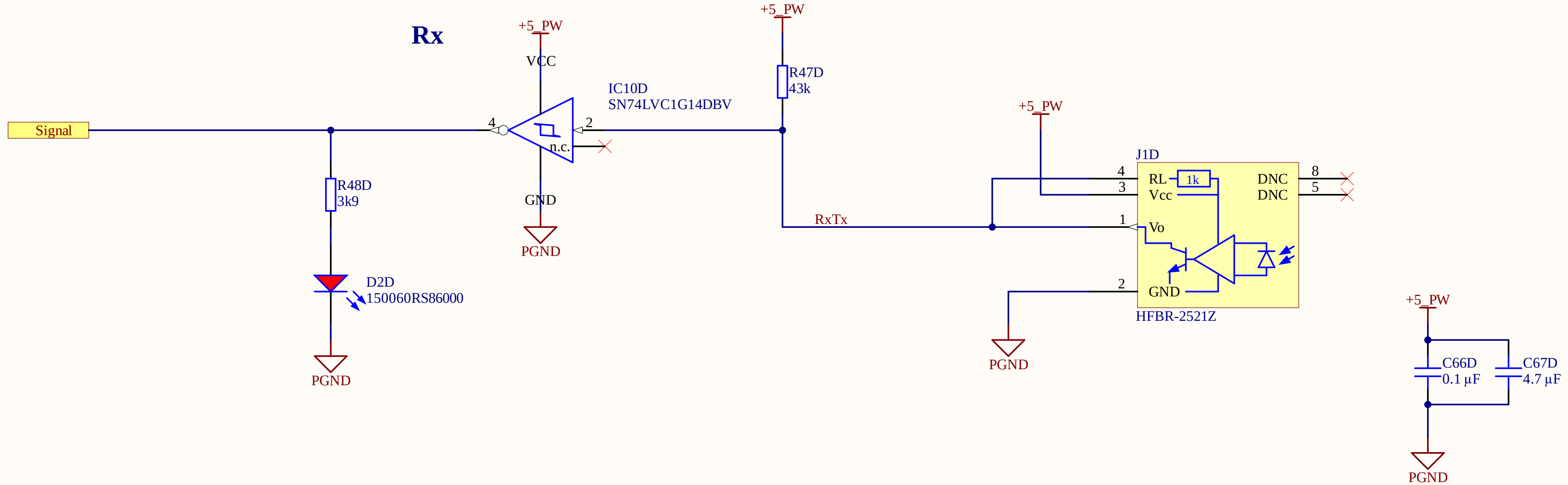
Title Voltage_Measurement.SchDoc		UltraZohm
Revision: Rev01	Design Engineer: Veit Starost	www.ultrazohm.com
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB		Date: 08.03.2023
		Sheet .3 of 25

Poor Mans Table of Truth for Tx

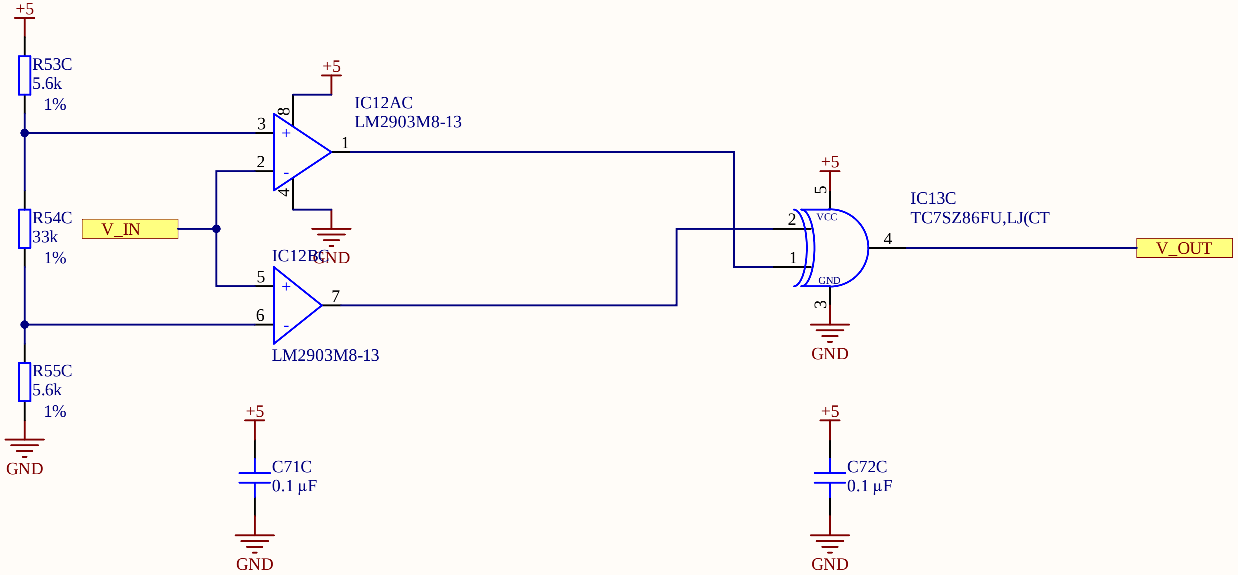
Transmitter LED	Node Tx_N	LED D3	Signal
OFF	HIGH	OFF	LOW
ON	LOW	ON	HIGH

Poor Mans Table of Truth for Rx

Receiver LED	Node RxTx	LED D2
OFF	HIGH	OFF
ON	LOW	ON



Title: Optical_RX.SchDoc	**UltraZohm**
Revision: Rev01 — Design Engineer: Veit Starost	www.ultrazohm.com
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB	Date: 08.03.2023 — Sheet .3 of 25



Window comparator circuit
Threshold calculation acc to screenshot
XOR to select areas where VIN is out of the window

VH 4.35V 45A
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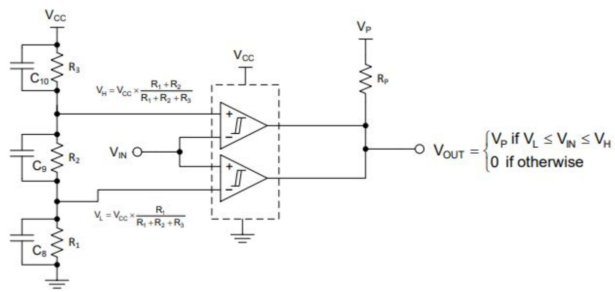

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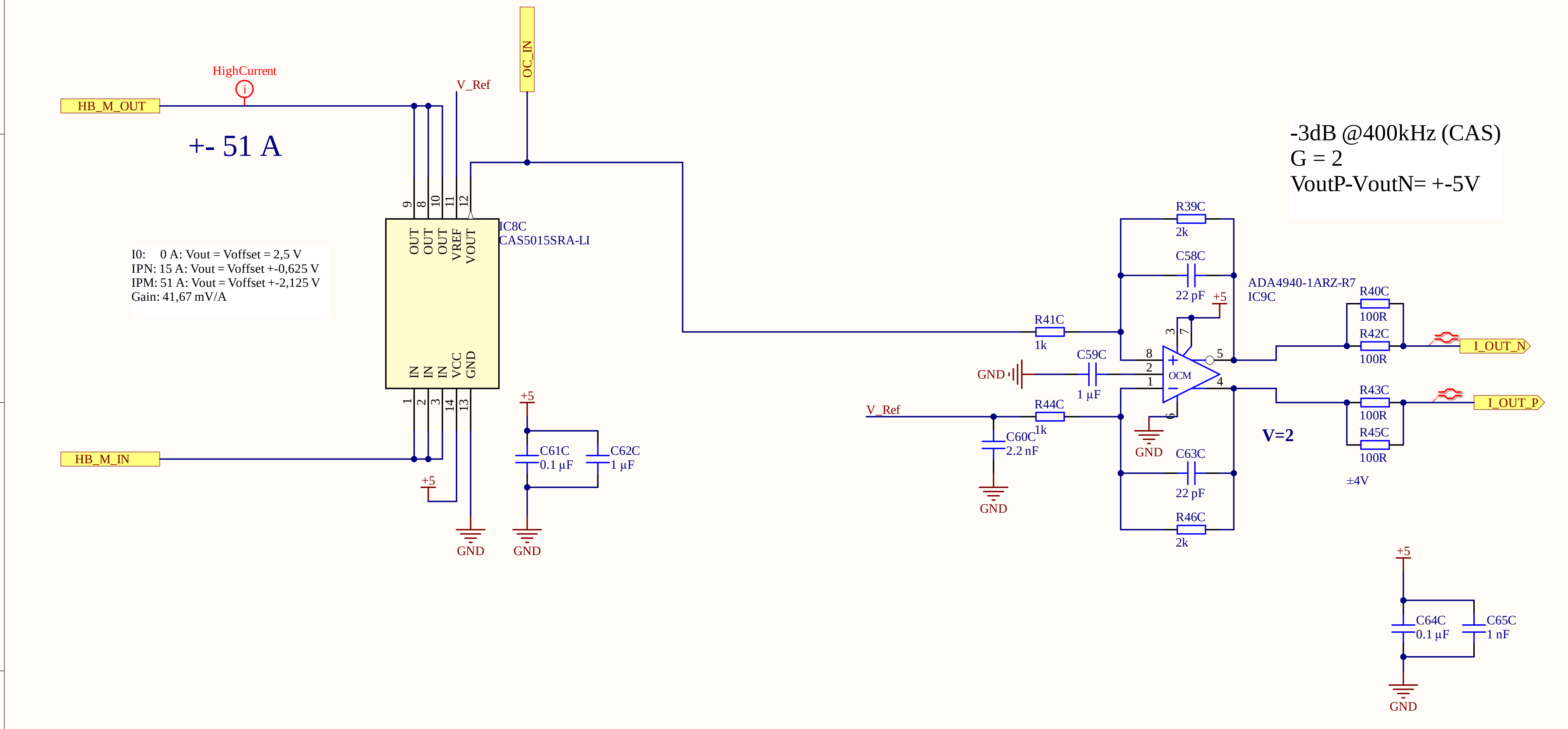
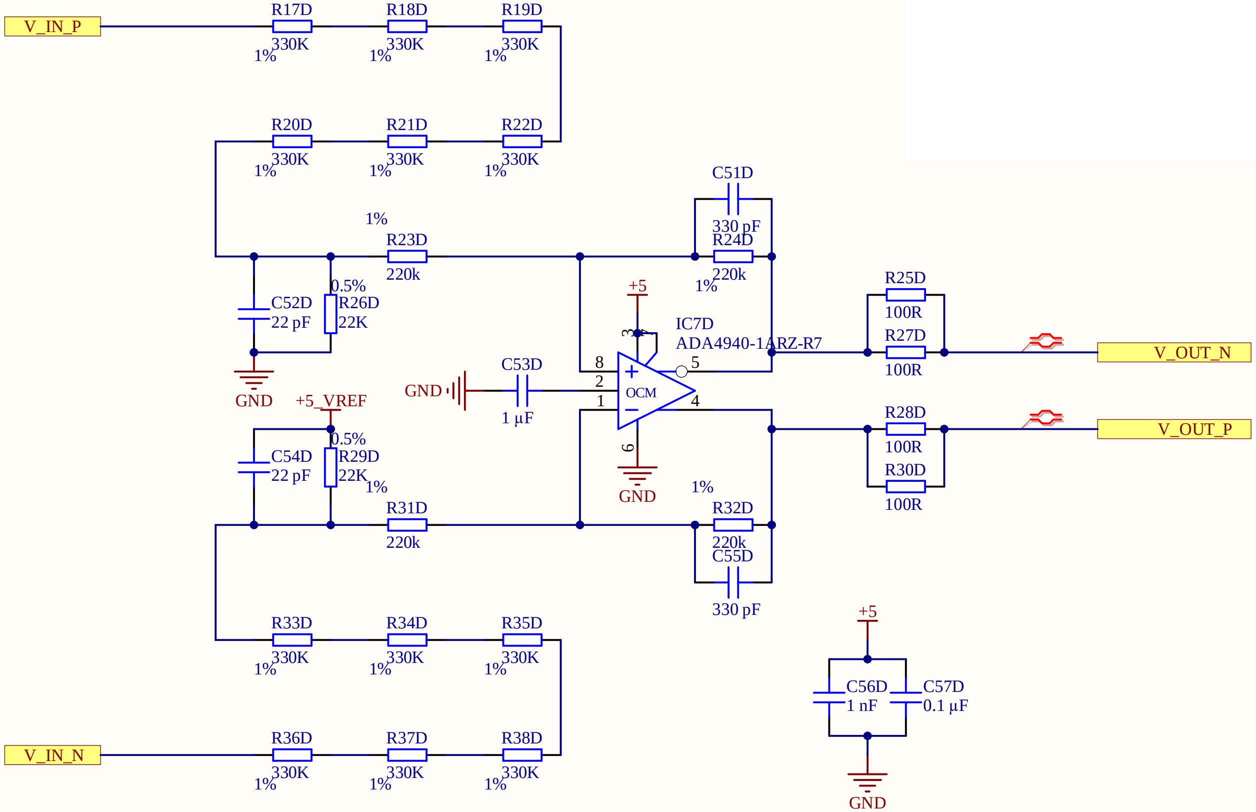
Title OC_Trigger.SchDoc		UltraZohm
Revision: Rev01	Design Engineer: Veit Starost	www.ultrazohm.com
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB		Date: 08.03.2023 Sheet .3 of 25



Title Current_Measurement_CAS5000.SchDoc		UltraZohm www.ultrazohm.com
Revision: Rev01	Design Engineer: Veit Starost	
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB		Date: 08.03.2023 Sheet .3 of 25

Vin: 0V - 900V
Vout: -4.5V - 4.5V
-3dB @2.2kHz
G = 0,01



Title Voltage_Measurement.SchDoc		**UltraZohm** www.ultrazohm.com
Revision: Rev01	Design Engineer: Veit Starost	Date: 08.03.2023
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB		Sheet .4 of 25

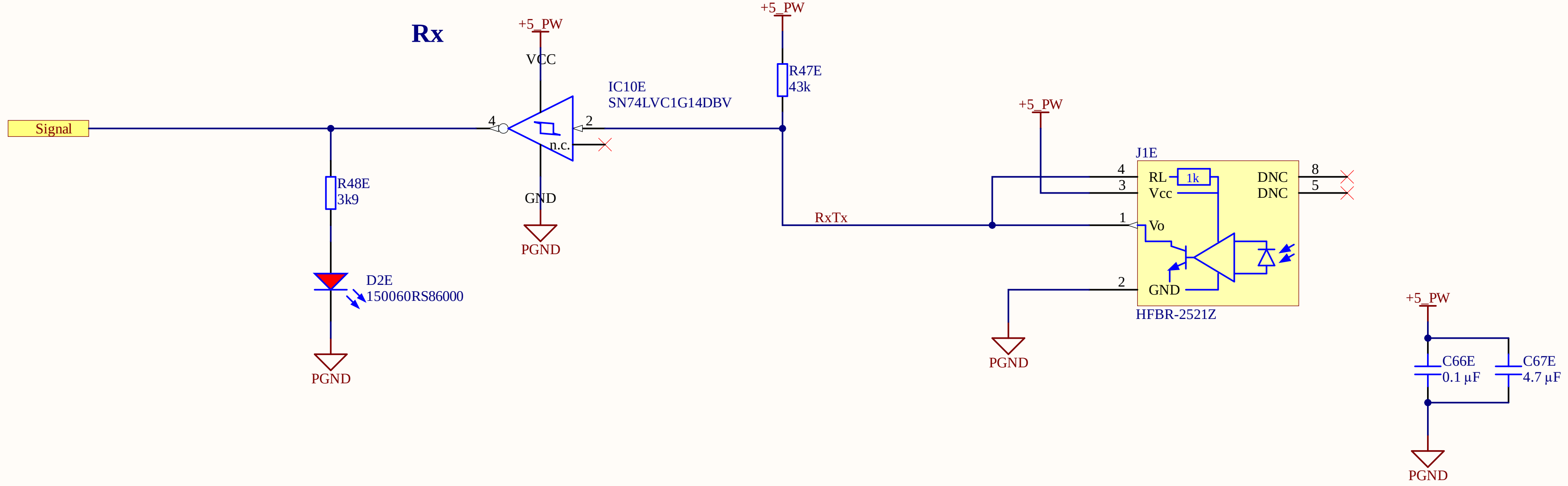
Poor Mans Table of Truth for Tx

Transmitter LED	Node Tx_N	LED D3	Signal
OFF	HIGH	OFF	LOW
ON	LOW	ON	HIGH

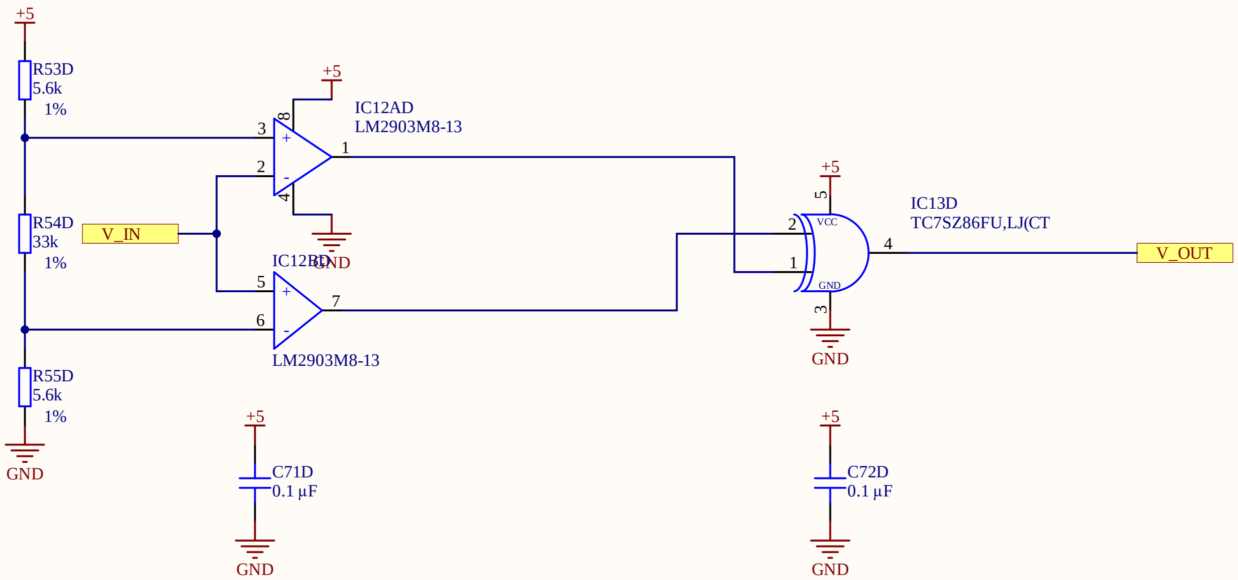
Poor Mans Table of Truth for Rx

Receiver LED	Node RxTx	LED D2
OFF	HIGH	OFF
ON	LOW	ON

Rx



Title Optical_RX.SchDoc		*UltraZohm*
Revision: Rev01	Design Engineer: Veit Starost	www.ultrazohm.com
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB		Date: 08.03.2023 Sheet .4 of 25



Window comparator circuit
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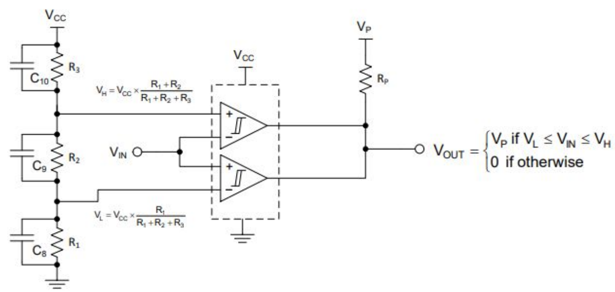

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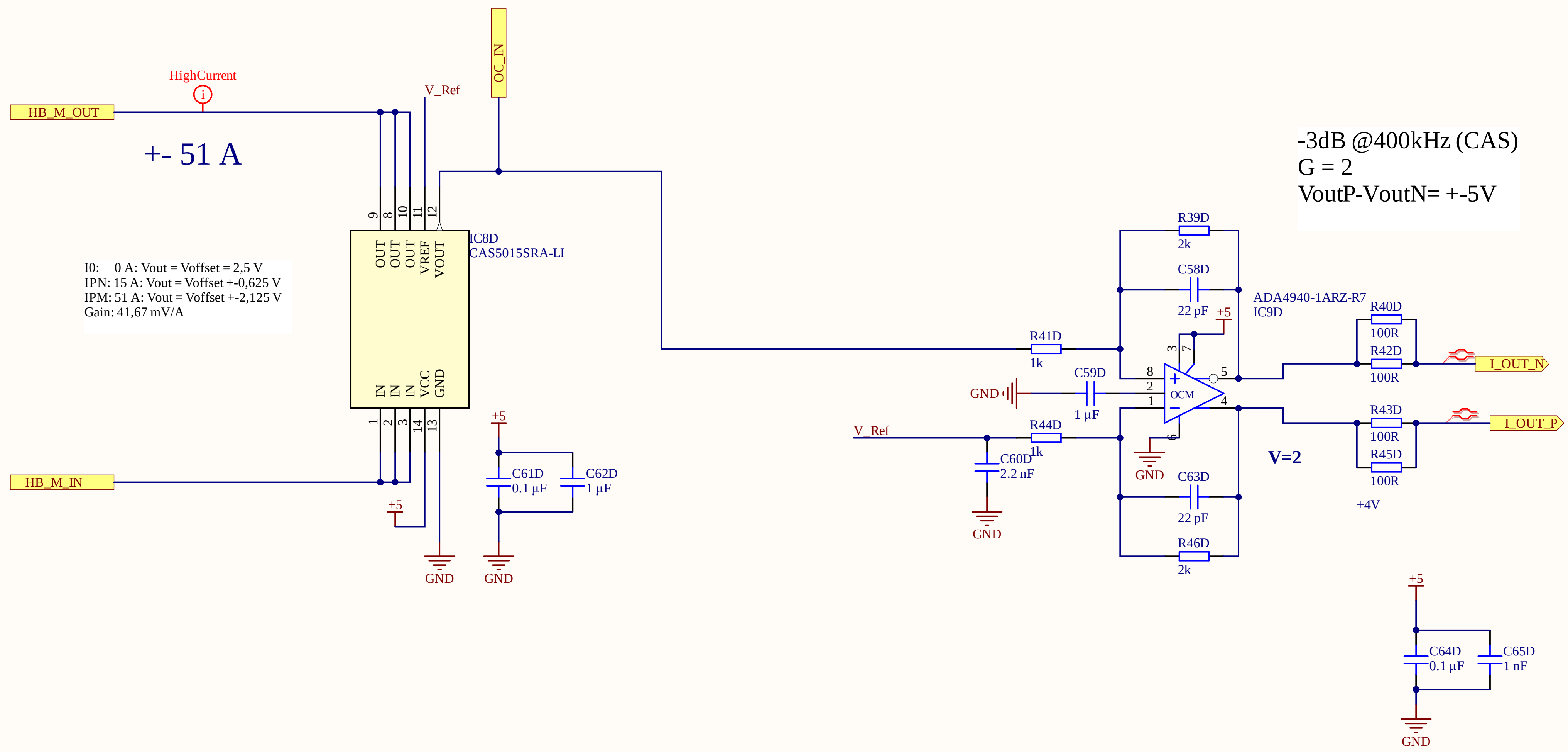
Title OC_Trigger.SchDoc		UltraZohm www.ultrazohm.com
Revision: Rev01	Design Engineer: Veit Starost	
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB		Date: 08.03.2023 Sheet .4 of 25



Title	Current_Measurement_CAS5000.SchDoc
Revision: Rev01	Design Engineer: Veit Starost
Project:	UZ_PER_SiC_inverter_828xcc.PrjPCB

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Date: 08.03.2023
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Poor Mans Table of Truth for Tx

Transmitter LED	Node Tx_N	LED D3	Signal
OFF	HIGH	OFF	LOW
ON	LOW	ON	HIGH

Poor Mans Table of Truth for Rx

Receiver LED	Node RxTx	LED D2
OFF	HIGH	OFF
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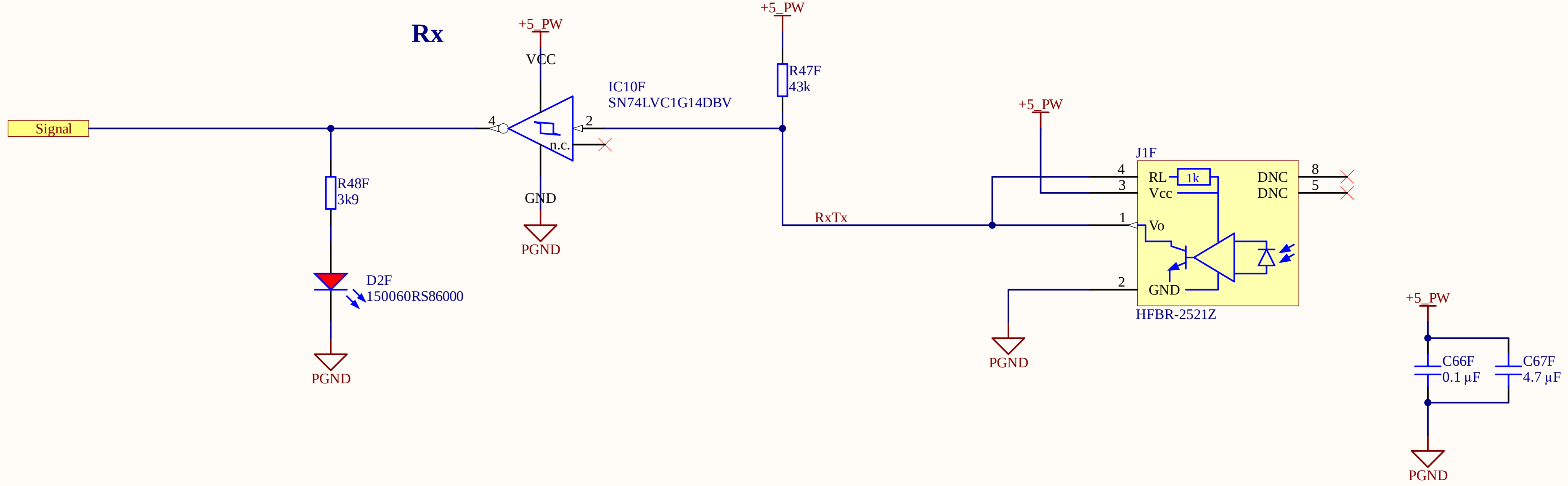


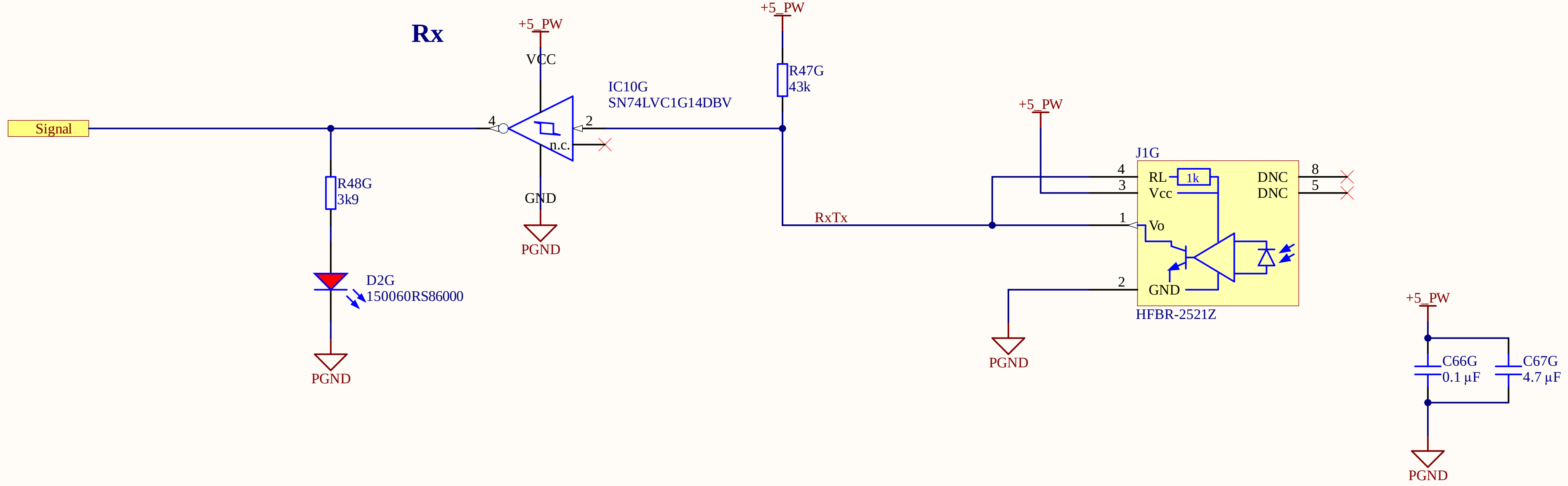
Title Optical_RX.SchDoc		UltraZohm	
Revision: Rev01	Design Engineer: Veit Starost	www.ultrazohm.com	
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB		Date: 08.03.2023 Sheet .5 of 25	

Poor Mans Table of Truth for Tx

Transmitter LED	Node Tx_N	LED D3	Signal
OFF	HIGH	OFF	LOW
ON	LOW	ON	HIGH

Poor Mans Table of Truth for Rx

Receiver LED	Node RxTx	LED D2
OFF	HIGH	OFF
ON	LOW	ON



Title	Optical_RX.SchDoc
Revision: Rev01	Design Engineer: Veit Starost
Project:	UZ_PER_SiC_inverter_828xcc.PrjPCB

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Date: 08.03.2023
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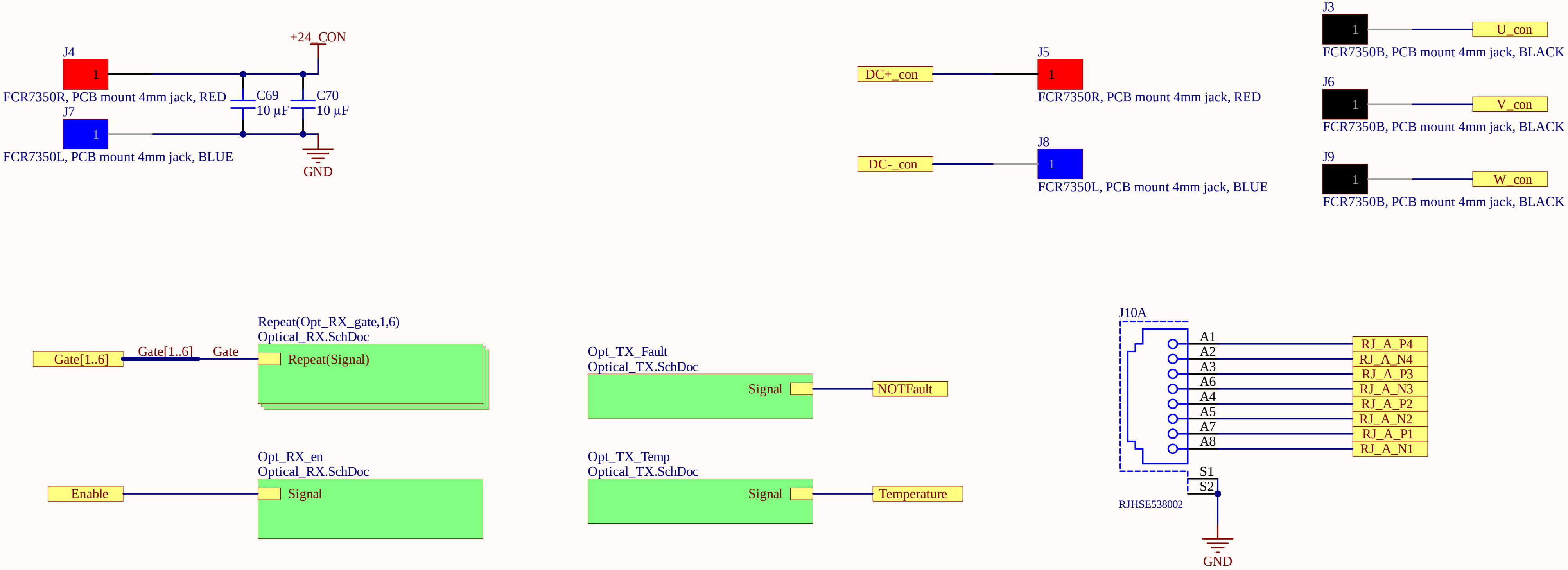

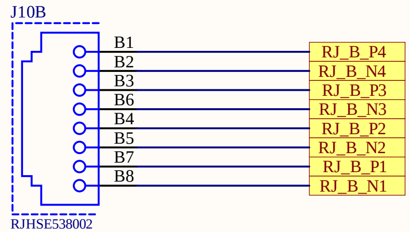

ADC Pinout	Signal	Name	ADC	Note
RJ45-A Connector J10A				
A6	i_ph_u_P	phase current U	ADC 3	inverse current
A3	i_ph_u_N			
A5	i_ph_v_P	phase current V	ADC 2	inverse current
A4	i_ph_v_N			
A8	i_ph_w_P	phase current W	ADC 1	inverse current
A7	i_ph_w_N			
A1	v_dc_P	dc link voltage	ADC 4	
A2	v_dc_N			
RJ45-B Connector J10B				
B6	v_ph_v_P	phase voltage V	ADC 3	
B3	v_ph_v_N			
B5	v_ph_w_P	phase voltage W	ADC 2	
B4	v_ph_w_N			
B8	i_dc_P	dc link current	ADC 1	
B7	i_dc_N			
B1	v_ph_u_P	phase voltage U	ADC 4	
B2	v_ph_u_N			

Title Connectors.SchDoc

Revision: Rev01 Design Engineer: Veit Starost

Project: UZ_PER_SiC_inverter_828xcc.PrjPCB

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Date: 08.03.2023
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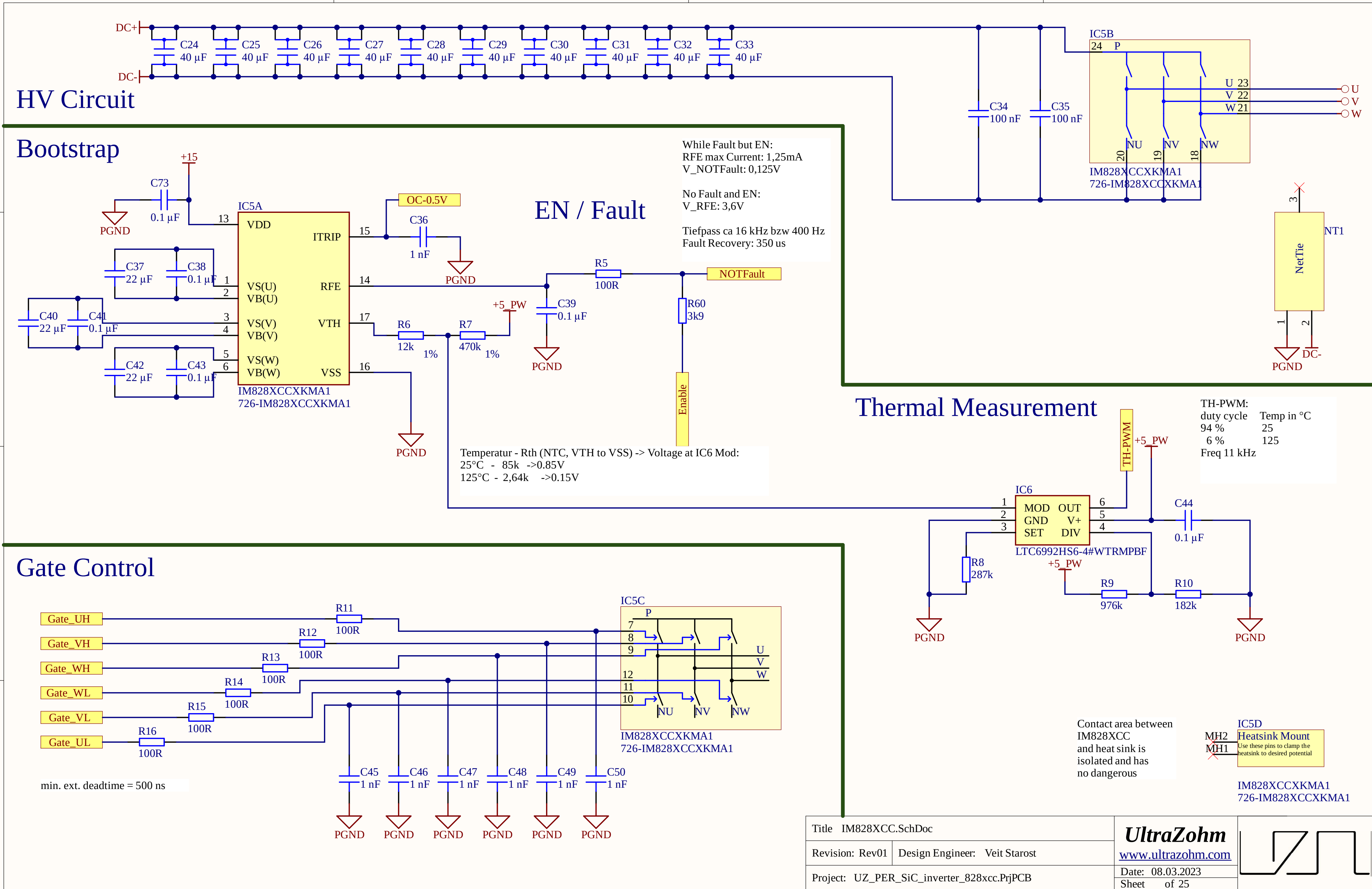
HV Circuit
DC+
DC-
C24 40 µF
C25 40 µF
C26 40 µF
C27 40 µF
C28 40 µF
C29 40 µF
C30 40 µF
C31 40 µF
C32 40 µF
C33 40 µF
C34 100 nF
C35 100 nF
IC5B
24 P
U 23
V 22
W 21
NU 20
NV 19
NW 18
U
V
W
IM828XCCXKMA1
726-IM828XCCXKMA1
Bootstrap
+15
C73 0.1 µF
PGND
IC5A
13 VDD
15 ITRIP
1 VS(U)
2 VB(U)
14 RFE
3 VS(V)
4 VB(V)
17 VTH
5 VS(W)
6 VB(W)
16 VSS
IM828XCCXKMA1
726-IM828XCCXKMA1
C37 22 µF
C38 0.1 µF
C40 22 µF
C41 0.1 µF
C42 22 µF
C43 0.1 µF
OC-0.5V
C36 1 nF
EN / Fault
R5 100R
NOTFault
C39 0.1 µF
R60 3k9
R6 12k 1%
R7 470k 1%
+5_PW
Enable
While Fault but EN:
RFE max Current: 1,25mA
V_NOTFault: 0,125V
No Fault and EN:
V_RFE: 3,6V
Tiefpass ca 16 kHz bzw 400 Hz
Fault Recovery: 350 us
Temperatur - Rth (NTC, VTH to VSS) -> Voltage at IC6 Mod:
25°C - 85k ->0.85V
125°C - 2,64k ->0.15V
NT1
NetTie
DC-
Thermal Measurement
TH-PWM
TH-PWM:
duty cycle Temp in °C
94 % 25
6 % 125
Freq 11 kHz
IC6
1 MOD
2 GND
3 SET
OUT 6
V+ 5
DIV 4
LTC6992HS6-4#WTRMPBF
C44 0.1 µF
R8 287k
R9 976k
R10 182k
Gate Control
Gate_UH
Gate_VH
Gate_WH
Gate_WL
Gate_VL
Gate_UL
R11 100R
R12 100R
R13 100R
R14 100R
R15 100R
R16 100R
IC5C
P
7
8
9
12
11
10
NU
NV
NW
IM828XCCXKMA1
726-IM828XCCXKMA1
C45 1 nF
C46 1 nF
C47 1 nF
C48 1 nF
C49 1 nF
C50 1 nF
min. ext. deadtime = 500 ns
Contact area between IM828XCC and heat sink is isolated and has no dangerous
IC5D
MH2
MH1
Heatsink Mount
Use these pins to clamp the heatsink to desired potential
IM828XCCXKMA1
726-IM828XCCXKMA1
Title IM828XCC.SchDoc
Revision: Rev01
Design Engineer: Veit Starost
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB
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www.ultrazohm.com
Date: 08.03.2023
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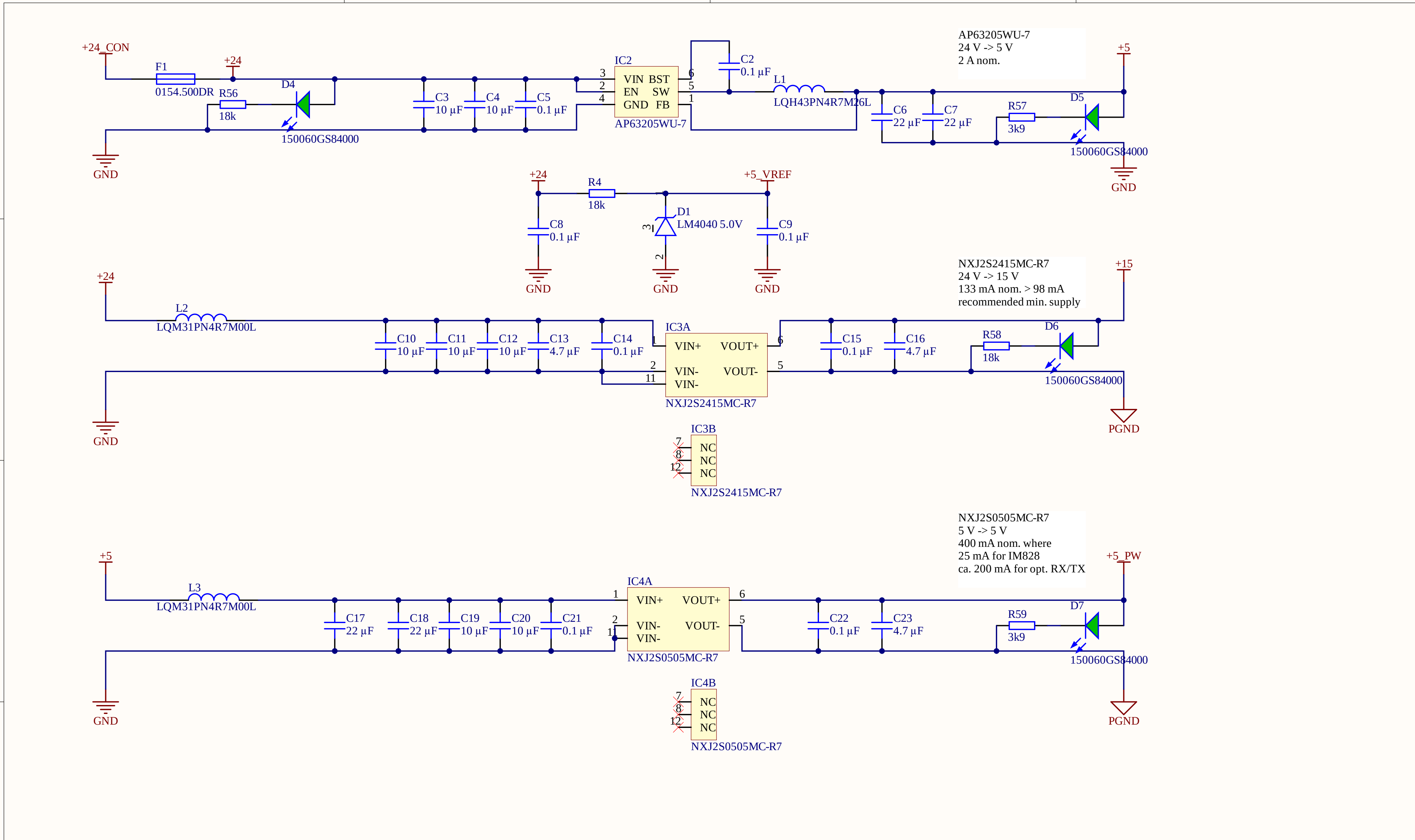

Title	Power.SchDoc
Revision: Rev01	Design Engineer: Veit Starost
Project: UZ_PER_SiC_inverter_828xcc.PrjPCB	

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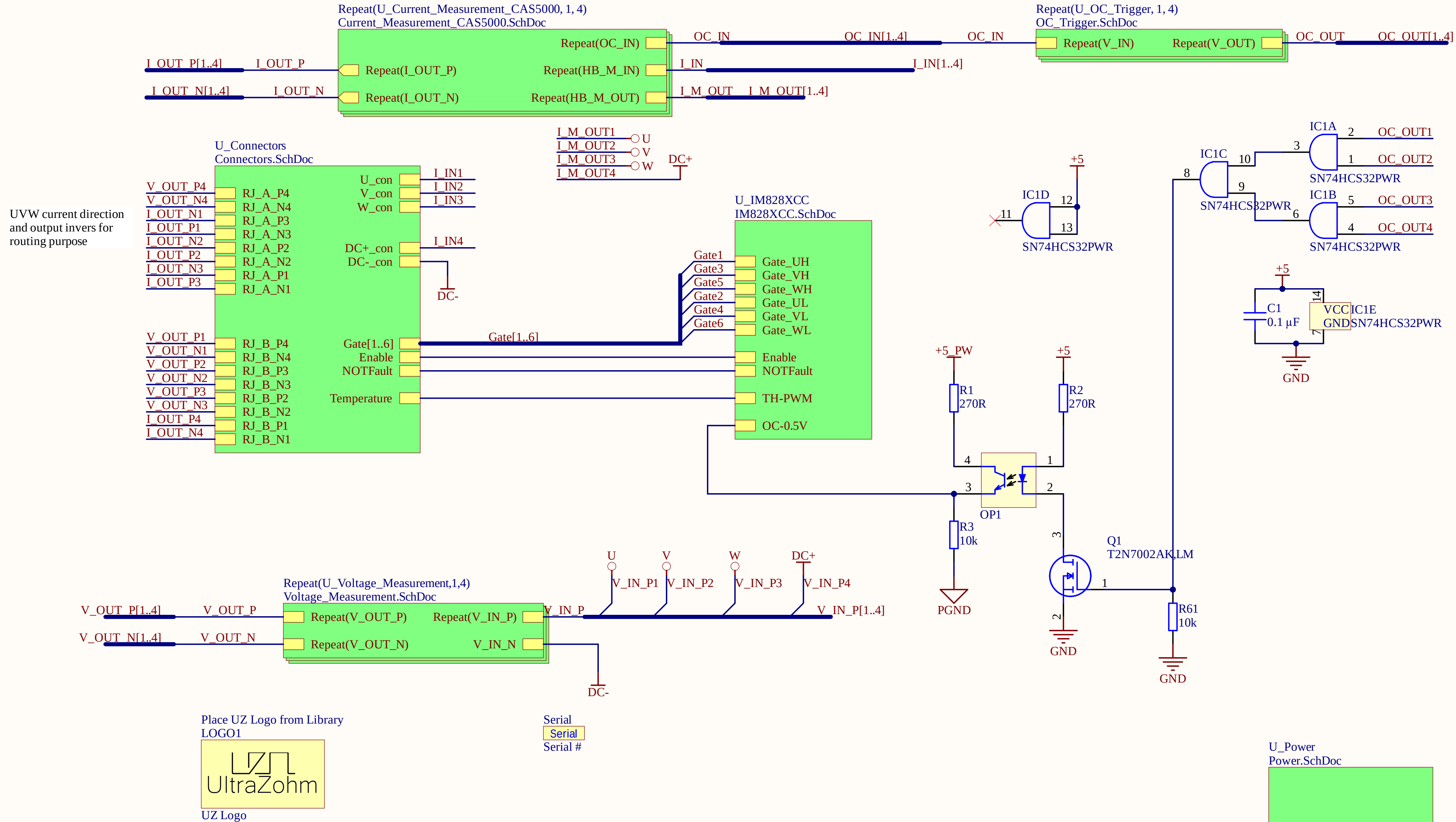

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Revision: Rev01	Design Engineer: Veit Starost	www.ultrazohm.com
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